Problem 15 | Problems T4_PhaseDiagrams I.pdf #15 | Ti-B · peritectics

The Ti-B System and Ti-10 % B and Ti-18 % B Alloys

Original statement · Problems T4_PhaseDiagrams I.pdf, p. 7

15. Given the phase diagram of the figure, answer the following questions: a. Write down the invariant reactions, indicating the name, temperature and composition of the phases involved. b. List the congruent and incongruent melting phases or compounds. c. For an alloy with 10% B, indicate the composition of the phases, the amount of phases and the amount of microconstituents at 1000 °C and 500 °C. d. Describe the solidification sequence of an alloy with 18% B at a very slow cooling from the liquid state up to 500 °C.

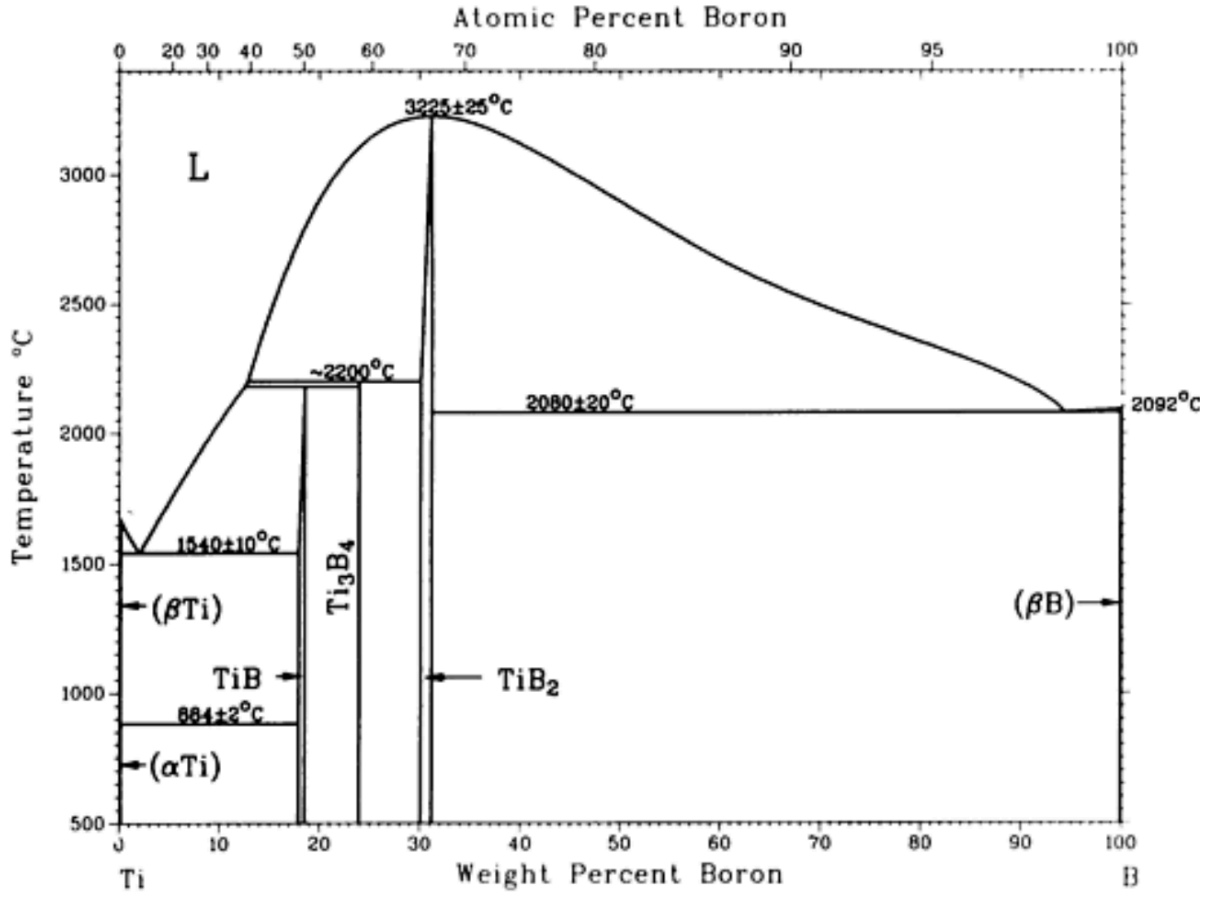


Figure 25. Ti-B phase diagram in weight percent (Problems T4_PhaseDiagrams I.pdf, p. 7).

Official Solution: a) peritectic (2200 °C); peritectic (2190 °C); eutectic (1540 °C); eutectic (2080 °C). b) Ti_3B_4 → L + TiB_2. c) 1000 °C: $C_{(\beta Ti)}$ = 0%B; C_{TiB} = 18%B; $X_{(\beta Ti)}$ = 0.44; $X_{(TiB)}$ = 0.56; eutectic (x = 50%) and primary TiB (x = 50%). 500 °C: $C_{(\alpha Ti)}$ = 0%B; C_{TiB} = 18%B; $X_{(\alpha Ti)}$ = 0.44; $X_{(TiB)}$ = 0.56; eutectic (x = 50%) and primary TiB (x = 50%)

PHASE 1 · HYPOTHESES & PARAMETERS

Hypotheses

1. **Equilibrium**, Ti-B diagram of the sheet, read on its **weight percent** scale (bottom axis). Printed values: 1540, 884, ~2200, 2080, 3225 and 2092 °C. Compositions are **readings of the figure** (±1 wt%).
2. Boron is practically insoluble in Ti: (βTi) and (αTi) are taken as 0 wt% B, as the key does.
3. TiB, Ti_3B_4 and TiB_2 are compounds (Slide 15 of Session 8 shows the same system on the atomic scale).

Data read on the diagram

Line	Phases (wt% B)
1540 °C eutectic	(βTi) ≈ 0, L ≈ 2, TiB ≈ 18
~2200 °C	L ≈ 13, Ti_3B_4 ≈ 24, TiB_2 ≈ 31
~2190 °C	L ≈ 13, TiB ≈ 18, Ti_3B_4 ≈ 24
2080 °C eutectic	TiB_2 ≈ 31, L ≈ 94, (βB) ≈ 100

PHASE 2 · FRAMES & STRATEGY

Frame: the Ti-B diagram (wt% B, °C); horizontal tie lines and the vertical line of each alloy; no change of basis. Names of reactions from their shape [Session 8 Slide 6]; amounts from the lever rule [Session 7 Slides 16, 32].

PHASE 3 · STEP-BY-STEP DERIVATION

a) Invariant reactions (on cooling)

T (°C)	Reaction	Name
~2200	$L\,(13) + \mathrm{TiB_2}\,(31) \rightarrow \mathrm{Ti_3B_4}\,(24)$	peritectic
~2190	$L\,(13) + \mathrm{Ti_3B_4}\,(24) \rightarrow \mathrm{TiB}\,(18)$	peritectic
2080	$L\,(94) \rightarrow \mathrm{TiB_2}\,(31) + (\beta\mathrm{B})\,(100)$	eutectic
1540	$L\,(2) \rightarrow (\beta\mathrm{Ti})\,(0) + \mathrm{TiB}\,(18)$	eutectic

At 884 °C the titanium changes from (βTi) to (αTi) (its allotropic change, Topic 2); TiB does not take part.

b) Melting

- **TiB_2 melts congruently**, at the top of the liquidus, 3225 °C.

- **Ti_3B_4 melts incongruently** ($Ti_3B_4 \rightarrow L + TiB_2$ on heating at ~2200 °C), and so does **TiB** ($TiB \rightarrow L + Ti_3B_4$ at ~2190 °C): both end at peritectic lines [Session 8 Slide 17].

c) Ti-10 wt% B at 1000 °C and 500 °C Phases at 1000 °C: (βTi) 0 and TiB 18 (length 18):

$$X_{(\beta\text{Ti})} = \frac{18-10}{18-0} = \frac{8}{18} = 0.44 \qquad X_{\text{TiB}} = \frac{10-0}{18} = \frac{10}{18} = 0.56$$

At 500 °C the same, with (αTi) instead of (βTi): 0.44 and 0.56.

Microconstituents. $2 < 10 < 18$: hypereutectic, primary TiB and eutectic. Short tie line from the eutectic liquid (2) to TiB (18), length 16:

$$X_{\text{TiB},p} = \frac{10-2}{18-2} = \frac{8}{16} = 0.50 \qquad X_E = \frac{18-10}{16} = 0.50$$

The same at 1000 and 500 °C (the 884 °C change affects only the Ti inside the eutectic).

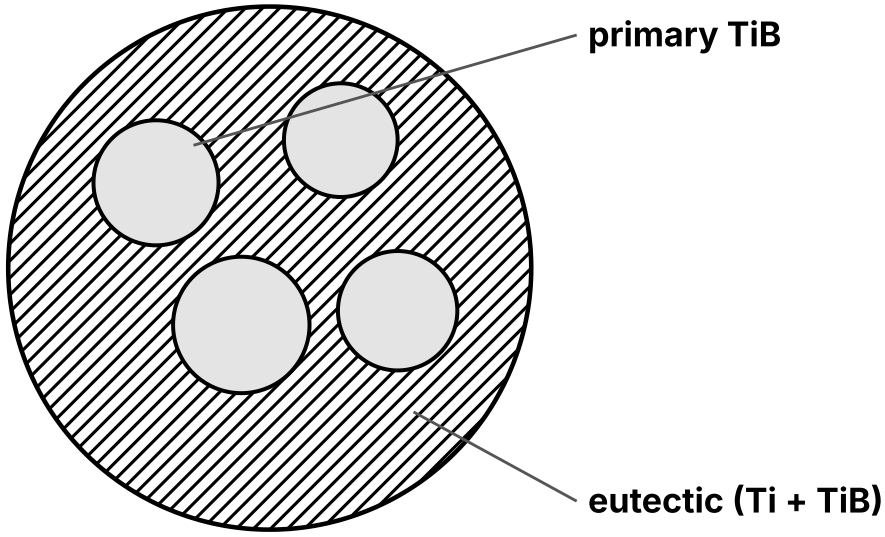


Figure 24. Ti-10 wt% B below 1540 °C: primary TiB (50 %) in the eutectic Ti + TiB (50 %).

d) Ti-18 wt% B, very slow cooling to 500 °C

1. **Liquid**, down to the liquidus on the TiB_2 side of the dome (roughly 2500 °C at 18 wt% B, a reading).
2. **L + TiB_2**: primary TiB_2 crystallises; the liquid becomes poorer in B, down to 13 wt% at ~2200 °C.
3. **~2200 °C, peritectic** $L + TiB_2 \rightarrow Ti_3B_4$: since $13 < 18 < 24$, all the TiB_2 is used up and the alloy is L + Ti_3B_4.
4. **~2190 °C, peritectic** $L + Ti_3B_4 \rightarrow TiB$: the alloy has exactly the composition of TiB, so liquid and Ti_3B_4 are both used up: **100 % TiB**.

5. From ~2190 to 500 °C nothing changes: the alloy stays single-phase TiB.

PHASE 4 · CHECKS & INTERPRETATION

Checks

- (c): $0.44 \times 0 + 0.56 \times 18 = 10$ wt% B; the eutectic (0.50) contains all the Ti (0.44) and $0.56 - 0.50 = 0.06$ of TiB, $0.44 + 0.06 = 0.50$.
- (d): in equilibrium each peritectic is complete; with faster cooling the TiB would surround the Ti_3B_4 (and that the TiB_2) and the reactions would stay incomplete [Session 8 Slide 10].

Interpretation A Ti-B alloy with 10 % B is half eutectic (Ti + TiB) and half primary TiB, a hard boride; the 18 % B alloy is a single compound, reached through two peritectic reactions in a row.

Check against the official solution

Parts (a), (b) and (c) agree with the key (0.44/0.56 and 50 %/50 % at both temperatures). Part (b) of the key lists only Ti_3B_4; TiB is also incongruent and TiB_2 congruent, as shown above. The key gives no answer for part (d).